Does a mixed model keep a plant GWAS calibrated? A re-analysis of Atwell et al. (2010) with 8 times more SNPs

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1 Question

Atwell et al. (2010) ran genome-wide association studies (GWAS) on 107 phenotypes in 199 *Arabidopsis thaliana* inbred lines. They report that a simple test (Wilcoxon) is badly confounded by population structure, that the EMMA mixed model removes the excess of association for flowering-related traits, and also that EMMA's p-values "are not always well estimated and should be interpreted with caution". This report re-asks three narrow questions with a different implementation and a much denser SNP set:

1. Does the pattern replicate: strong inflation without correction, near-null calibration with a mixed model?
2. Does a leave-one-chromosome-out (LOCO) kinship, which avoids using a SNP's own region to model relatedness (Yang et al., 2014), change calibration or hits compared with an all-SNP kinship?
3. Do four well-known loci, chosen in advance, localise with the denser SNPs: *FLC*, *FRI* and *FT* for flowering traits, and *DOG1* for seed dormancy?

This is not an attempt to "beat" the paper, which already used a mixed model; it is a check on calibration and localisation with different data and software.

2 Data

Phenotypes. AraPheno study 1 (the Atwell et al. phenotypes), 107 phenotypes with 51 to 135 accessions each (median 92) after matching to genotypes. Replicate rows were averaged per accession, and each phenotype was rank-based inverse-normal transformed so that a parametric test is comparable with the paper's rank-based Wilcoxon test.

Genotypes. The 1001 Genomes imputed binary SNP matrix (1001 Genomes Consortium, 2016): 10,709,949 SNPs for 1,135 accessions. **Only 137 of the 199 lines in the original paper are in this release**, so these analyses have less power than the original. SNPs with minor-allele frequency below 5% among those 137 accessions were dropped, leaving 1,826,193 SNPs, about 8 times the 216,130 SNPs in the original study. I found no licence statement for the genotype matrix, so the repository contains download scripts, not data.

3 Methods

All code is in `src/`; the statistical core (`src/lmm.py`) has tests (`tests/`) that check it against `scipy`'s OLS and against a brute-force generalised-least-squares computation.

- **OLS:** simple linear regression of the transformed phenotype on each SNP, no correction.
- **Mixed model (all-SNP kinship), EMMAX-style (Kang et al., 2010):** $y = \mu + x\beta + g + e$ with $g \sim N(0, \sigma_g^2 K)$. K is a realised relationship matrix from every 20th SNP (about 91,000 SNPs). The variance ratio $\delta = \sigma_g^2/\sigma_e^2$ is estimated once per phenotype by REML under the null and then fixed for all SNPs (the "P3D" approximation), after which each SNP needs a closed-form weighted regression.
- **Mixed model with LOCO kinship:** the same, but each chromosome is scanned with a kinship computed from SNPs on the other four chromosomes.

- **Significance:** Bonferroni, $0.05/1{,}826{,}193 = 2.7 \times 10^{-8}$. This is conservative because nearby SNPs are in strong linkage disequilibrium (LD).
- **Calibration:** genomic inflation factor λ (median χ^2 over 0.455) per phenotype and method.
- **Candidate loci, fixed before looking at results:** for each pre-specified trait–gene pair (12 flowering traits against *FLC*, *FRI*, *FT*; 2 dormancy traits against *DOG1*; 38 pairs) I take the smallest p-value within the gene plus or minus 50 kb. Because that window holds about a thousand SNPs, a small minimum is unremarkable, so each is compared with 2,000 random windows of the same width: the **window empirical p** is the share of random windows with an equal or smaller minimum.

Software. Python 3.12, NumPy, SciPy, h5py; a fixed random seed for the null windows.

4 Results

4.1 Calibration: the paper's qualitative finding replicates

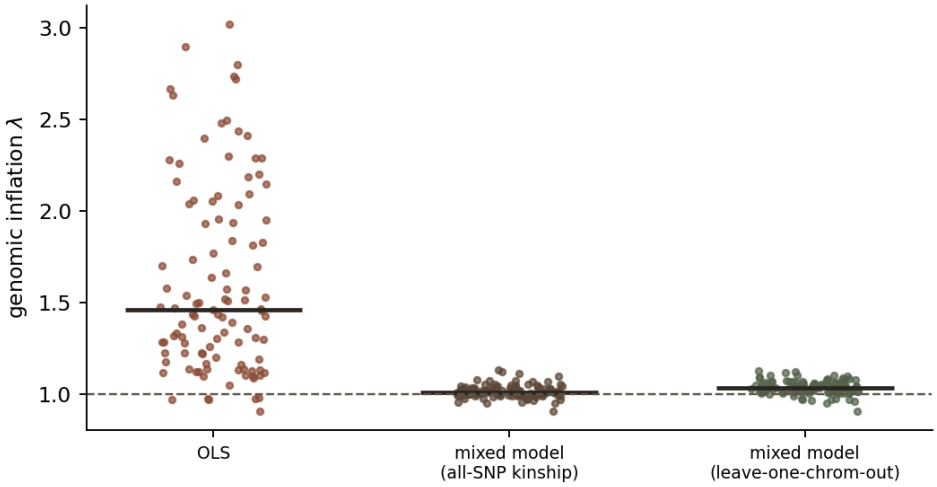


Figure 1: Genomic inflation factor for each of the 107 phenotypes under three methods. Bars: medians. Dashed line: $\lambda = 1$ (well calibrated).

Without correction, 76% of phenotypes have $\lambda > 1.2$ (median 1.46, interquartile range 1.21 to 2.00, maximum 3.02). With a mixed model the inflation is gone: median 1.01 with the all-SNP kinship and 1.04 with LOCO kinship, and no phenotype exceeds $\lambda = 1.13$ (Figure 1). This matches what Atwell et al. (2010) describe for flowering-related traits, here across all phenotypes and with a different implementation.

4.2 What this does to the number of "hits"

Table 1: Bonferroni-significant SNPs ($p < 2.7 \times 10^{-8}$) across the 107 phenotypes. SNP counts overstate independent signals because neighbouring SNPs are in LD; the last column is a rough count of independent regions.

Method	Phenotypes with ≥ 1 hit	Hit SNPs	Distinct 200 kb regions
OLS	52	48,631	2,156
Mixed model, all-SNP kinship	15	2,314	212
Mixed model, LOCO kinship	16	2,260	211

37 phenotypes have at least one OLS hit and no LOCO mixed-model hit; only 1 has the reverse. Flowering time

shows it clearly: for FT16, OLS gives $\lambda = 3.0$ and 10,737 significant SNPs spread across all five chromosomes, while the LOCO mixed model gives one (Figure 2).

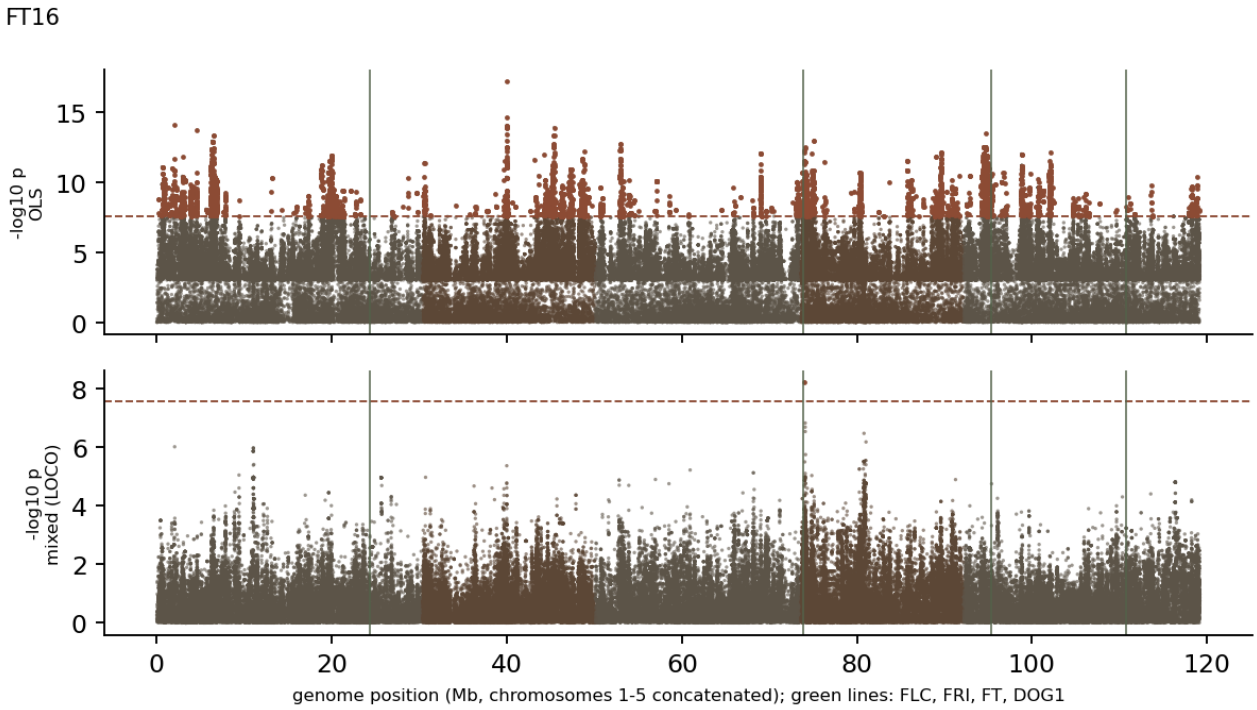


Figure 2: FT16 (n = 134), unadjusted (top) and mixed model with LOCO kinship (bottom). Dashed line: Bonferroni threshold. Green lines: FLC, FRI, FT, DOG1.

The single mixed-model hit for FT16 lies at chromosome 4, position 435,252. It looks adjacent to *FRI* on the concatenated axis, but *FRI* ends at 271,503, so the hit is about 164 kb away and outside the pre-specified window. I do not count it as localising on *FRI*.

Phenotypes with the most mixed-model hits are not flowering traits and have small samples (for example YEL, n = 64, 1,792 LOCO hit SNPs). I did not investigate them; many such SNPs are likely to be in LD with one or a few lead variants.

4.3 All-SNP versus leave-one-chromosome-out kinship

The two mixed models give almost the same hits (2,314 against 2,260 SNPs, 212 against 211 regions). Calibration differs slightly: LOCO has the higher median λ (1.04 against 1.01). A paired Wilcoxon test makes that difference highly detectable ($p = 1.2 \times 10^{-17}$) but it is small in size. It is consistent with the all-SNP kinship slightly deflating test statistics because it includes SNPs near the one being tested, which LOCO is designed to avoid (Yang et al., 2014); I did not test that mechanism directly. For these data the choice does not change the conclusions.

4.4 Do the known loci localise?

Table 2: Pre-specified trait-locus pairs with window empirical p < 0.05 under the LOCO mixed model.

Trait	Locus	OLS	Mixed (all-SNP)	Mixed (LOCO)
FT10	FLC	0.038	0.001	0.002
FT Field	FT	0.412	0.020	0.016
0W GH FT	FT	0.511	0.014	0.018
FT16	FLC	0.066	0.012	0.019

Trait	Locus	OLS	Mixed (all-SNP)	Mixed (LOCO)
LN22	FLC	0.100	0.042	0.030
LN22	FRI	0.080	0.034	0.037
SDV	FLC	0.288	0.051	0.040
LN16	FT	0.653	0.034	0.041

Of the 38 pre-specified pairs, 8 have a window empirical $p < 0.05$ under the LOCO mixed model, against about 1.9 expected by chance; under OLS only 2 do. Four of the eight are flowering traits at *FLC* (FT10, FT16, LN22, SDV), three are at *FT* (0W GH FT, FT Field, LN16) and one is at *FRI* (LN22, which also appears at *FLC*); FT22 at *FLC* just misses under LOCO (0.054); none involves *DOG1* (seed-dormancy windows: empirical p from 0.71 to 0.89), and the *FLC* expression phenotype does not localise on *FLC* or *FRI* (empirical p 0.39 and 0.17).

Two cautions apply. First, the flowering phenotypes are strongly correlated with each other (Atwell et al., 2010), so these eight pairs are not eight independent confirmations. Second, after a Bonferroni correction over 38 pairs (threshold 0.0013), only FT10–*FLC* under the all-SNP kinship ($p = 0.001$, borderline) would survive; under LOCO it is 0.003, which does not. The evidence is suggestive for *FLC* and *FT* in flowering traits, and absent for *DOG1* at this sample size.

4.5 Heritability estimates

The REML estimates of $\sigma_g^2/(\sigma_g^2 + \sigma_e^2)$ have a median of 0.72, but 18% of phenotypes sit at the upper boundary (> 0.99) and 5% at the lower (< 0.01). With 51 to 135 lines these estimates are imprecise (my own simulation test gave a standard deviation of about 0.3 at $n = 200$), so I treat them only as an input to the mixed model, not as findings.

5 Discussion and limitations

What this shows. On 1.83 million SNPs and 137 of the original lines, an independent mixed-model implementation reproduces the paper's calibration result across all 107 phenotypes, and a leave-one-chromosome-out kinship changes little. Mixed models also recover more flowering-gene signals than uncorrected tests (8 against 2 pairs below 0.05), though the evidence is modest.

What it does not show. It does not improve on the paper's conclusions, and I did not compare my hit lists with the paper's supplementary tables, so I cannot say whether the same loci are found. The data overlap the original study (the same phenotypes and 137 of the same lines), so this is a robustness check, not an independent replication.

- **Power.** Only 137 of 199 lines are in the genotype release; sample sizes are 51 to 135. A non-significant result, such as for *DOG1*, is not evidence of no effect.
- **Method choices.** Phenotypes were rank-inverse-normal transformed; the paper's choices may differ. The P3D approximation fixes the variance ratio per phenotype. Kinship uses about 91,000 thinned SNPs. Bonferroni correction over all SNPs is conservative under strong LD.
- **Genotypes.** The matrix is imputed, and imputation error is not modelled. SNPs were filtered at 5% minor-allele frequency among the 137 lines.
- **Candidate-gene test.** Pre-specified, but the pairs are correlated and the window (gene plus or minus 50 kb) differs from the paper's 20 kb windows around its top SNPs.
- **Data.** No licence statement was found for the 1001 Genomes genotype matrix, so it is not redistributed here; `src/00_download.sh` fetches it.

6 Provenance and reproducibility

All data are public (AraPheno, the 1001 Genomes, Ensembl Plants for gene coordinates). Run `src/00_download.sh`, then `src/01_prepare.py`, `src/02_scan.py` (about 40 minutes on a laptop) and `src/03_summarise.py`, then render this report. Tests: `python -m pytest`. The code and report were written

and run with the assistance of Claude Code (Anthropic); I am responsible for checking and understanding the results. No part derives from university coursework or assessed material.

References

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